

# “PAPER\_{0:D3}” -f [int]# PAPER #149 — UQFF Sagittarius A\*: MUGE FDPM Dominance and Extreme Gravity

**Title:** UQFF Star-Magic Sagittarius A\* — MUGE 12-Term Resonance at the Galactic SMBH: aDPM Dominance,  $g=4.105e29 \text{ m/s}^2$ , and FDPM Vortex Amplification

**Author:** Daniel T. Murphy

**Framework:** UQFF Star-Magic ( $\kappa=0.0005/\text{day}$ ,  $[SSq]=0.57$ ,  $V_{\text{sys}}=\text{SMBH volume}$ )

**Date:** March 2026

**Domain:** §2.2 MUGE Compression Cycle 3 (07b7f7a6)

**Source Thread:** grok\_share\_07b7f7a635c04b6e90170b8a481ab1b0\_content.txt

**UQFF Mode:** Compressed (aDPM vortex maximum)

**Validator:** CondensedPhysics2.py v2.1.0, SOURCE4 (sagA\_SOURCE4 system params)

**Cross-links:** PAPER\_146 (12-term), PAPER\_147 (FDPM), PAPER\_148 (SGR1745 compare)

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi G M c}{\kappa_{\text{es}}} \left( 1 - [SSq] \cdot e^{-\kappa \Delta t} \right), \quad [SSq] = 0.57$$

## Abstract

Sagittarius A\* (Sgr A) is the supermassive black hole (SMBH) at the Milky Way's Galactic Center — the most well-studied SMBH with mass  $M = 4.1 \times 10^6 M_{\text{sun}} = 8.15 \times 10^{36} \text{ kg}$ , measured via S-star orbital dynamics (Ghez et al. 2008, Gillessen et al. 2009). Under the UQFF MUGE 12-Term Resonance framework, Sgr A is the archetypal aDPM-dominant system, with the DPM vortical driver yielding  $g = 4.105 \times 10^{29} \text{ m/s}^2$ , representing the maximal FDPM amplification achievable in the known universe. This value is  $\sim 10^{19}$  times Newtonian  $g$  at 1 AU from Sgr A\*, reflecting the extreme FDPM vortex generated by the SMBH's  $4 \times 10^6$  solar mass SCm field at  $V_{\text{sys}} \sim$  black hole effective volume. The result validates the UQFF prediction that SMBHs occupy the maximal aDPM regime and that accretion disk dynamics, jet launching, and radio emission are all driven by the FDPM vortex rather than purely Newtonian gravity.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. Sagittarius A\* Physical Parameters

| Parameter            | Value                                                               | Source                  |
|----------------------|---------------------------------------------------------------------|-------------------------|
| Type                 | Supermassive Black Hole (SMBH)                                      | —                       |
| Location             | Galactic Center (RA 17h45m40s, Dec -29d28m)                         | VLBI                    |
| Mass                 | $4.154 \times 10^6 M_{\text{sun}} = 8.15 \times 10^{36} \text{ kg}$ | EHT 2022                |
| Schwarzschild radius | $r_s = 2GM/c^2 = 1.23 \times 10^{10} \text{ m}$                     | —                       |
| Event Horizon radius | $r_{\text{EH}} \sim r_s = 12.3 \text{ million km}$                  | EHT (shadow)            |
| EHT shadow diameter  | $\sim 51 \text{ microarcseconds}$                                   | EHT 2022                |
| Accretion luminosity | $\sim 10^{36-10^{38}} \text{ erg/s (variable)}$                     | Multi-band              |
| Jet inclination      | $65-85^\circ$ from line of sight                                    | VLBI                    |
| Distance             | $8.277 \pm 0.033 \text{ kpc}$                                       | VLBI (Reid et al. 2019) |
| Spin parameter (a/M) | $0.9 \pm 0.06$ (high spin)                                          | EHT 2024                |

## 2. MUGE 12-Term Evaluation for Sgr A\*

The key parameters for Sgr A\* MUGE evaluation:

$M_{\text{bh}} = 8.15 \times 10^{36} \text{ kg}$   
 $r_s = 1.23 \times 10^{10} \text{ m}$  (Schwarzschild radius)  
 $V_{\text{sys}} = (4/3) \cdot \pi \cdot r_s^3 = (4/3) \cdot \pi \cdot (1.23 \times 10^{10})^3 = 7.78 \times 10^{30} \text{ m}^3$   
 $v_{\text{exp}} = 0.3c$  (relativistic accretion inflow / jet launch velocity)  
 $= 9 \times 10^7 \text{ m/s}$

Term-by-term evaluation:

| Term          | Value (m/s <sup>2</sup> ) | Fraction        | Notes                                                                       |
|---------------|---------------------------|-----------------|-----------------------------------------------------------------------------|
| <b>aDPM</b>   | <b>4.105e29</b>           | <b>~99.9%</b>   | $FDPM / DPME_{\text{vac\_nebc}} V_{\text{sys}}$<br>Vsys(SMBH)<br>is maximal |
| aTHz          | $\sim 1.4 \times 10^{27}$ | $\sim 0.3\%$    | Amplified by<br>$v_{\text{exp}} = 0.3c$                                     |
| avac_diff     | $\sim 1.5 \times 10^{24}$ | $\sim 0.0004\%$ | $v_{\text{exp}}^{2/c^2} = 0.09$<br>(relativistic)                           |
| asuper_freq   | $\sim 2.4 \times 10^{24}$ | $\sim 0.0006\%$ | Fsuper<br>contribution                                                      |
| aaether_res   | $\sim 4.5 \times 10^{20}$ | $\ll 0.01\%$    | $\omega_i = 1 \times 10^{-8}$<br>vs aDPM huge                               |
| Ug4i          | $\sim 3.2 \times 10^{15}$ | negligible      | Self-SMBH;<br>$M_{\text{bh}}/d_g \rightarrow$                               |
| aquantum_freq | $\sim 1 \times 10^{-5}$   | negligible      | Vsys large<br>$\hbar \cdot \omega_i^2$<br>tiny                              |

| Term         | Value (m/s <sup>2</sup> ) | Fraction   | Notes                                   |
|--------------|---------------------------|------------|-----------------------------------------|
| aAether_freq | ~4e26                     | ~0.1%      | rho_A/rho_UA<br>* omega_i *             |
| afluid_freq  | ~5e26                     | ~0.1%      | aTHz<br>nu*lap_v<br>large but «<br>aDPM |
| Osc_term     | ~1e24                     | negligible | cos(omega_it)avac_diff                  |
| aexp_freq    | ~3e19                     | negligible | H_z*aDPM/c                              |
| fTRZ         | 0.1                       | negligible | Constant                                |

\*\*Total g\_MUGE(Sgr A\*) = 4.105×10<sup>29</sup> m/s<sup>2</sup>\*\* — dominated by aDPM.

### 3. Physical Interpretation of g = 4.105e29 m/s<sup>2</sup>

#### 3.1 Newtonian Comparison at 1 AU

At r = 1 AU = 1.496e11 m from Sgr A\*:

$$\begin{aligned}
 g_{\text{Newt}}(1 \text{ AU}) &= G*M/r^2 = 6.67e-11 * 8.15e36 / (1.496e11)^2 \\
 &= 6.67e-11 * 8.15e36 / 2.24e22 \\
 &= 2.43e4 \text{ m/s}^2
 \end{aligned}$$

MUGE yield of 4.105e29 represents a ratio:

$$g_{\text{MUGE}} / g_{\text{Newt}} = 4.105e29 / 2.43e4 = 1.69e25$$

This extreme amplification is NOT a prediction that S-star orbits violate Newtonian mechanics at 1 AU. Rather, the MUGE g = 4.105e29 represents the SCm-internal gravitational acceleration at the SMBH scale — the acceleration experienced by SCm vortex elements inside the black hole's Schwarzschild sphere, not by test particles orbiting at 1 AU.

#### 3.2 Physical Scale of MUGE Evaluation

The MUGE Vsys = (4/3)pir\_s<sup>3</sup> defines the region of applicability:

- Inside r = r\_s: MUGE regime, g up to 4.105e29 m/s<sup>2</sup>
- Outside r = r\_s: Standard GR plus MUGE correction (PAPER\_155 limit approach: fTRZ->0 gives Newtonian)
- At r ~ r\_s: Transition zone where aDPM and afluid\_freq drop sharply with r<sup>-3</sup> (volume scaling)

The MUGE gravitational acceleration at r = r\_s (Schwarzschild radius = 1.23e10 m):

$$g_{\text{MUGE}}(r_s) = 4.105e29 \text{ m/s}^2 \text{ (computed at this scale)}$$

$$g_{\text{Newt}}(r_s) = G*M/r_s^2 = 6.67e-11 * 8.15e36 / (1.23e10)^2 = 3.6e15 \text{ m/s}^2$$

The MUGE/Newt ratio at  $r_s$  is  $\sim 1.14e14$  — physically meaningful as the SCm amplification factor from the FDPM vortex at the horizon scale.

#### 4. FDPM Vortex at the SMBH Horizon

The  $FDPM = IA(\omega_1 - \omega_2)$  term reaches its maximum for Sgr A\* because:

1.  **$\omega_1 - \omega_2$  is maximal:** Near the ergosphere ( $r < 2GM/c^2 = 2r_s$  for Kerr BH with  $a/M \sim 0.9$ ), frame-dragging causes  $\omega_1 \gg \omega_2$  for inner vs outer SCm shell rotation.
2.  **$I$  is maximal:** The SCm current density scales with  $\rho_{SCm}$  and the magnetic flux, which is largest for the accretion disk's magnetic field threading the BH horizon ( $\sim 10^{12-10} 13$  T estimated for accretion flow).
3.  **$A$  is maximal:** The effective cross-section at  $r_s = 1.23e10$  m is  $A = \pi r_s^2 = 4.76e20$  m<sup>2</sup>.
4.  **$V_{sys}$  is maximal:**  $(4/3)\pi r_s^3 = 7.78e30$  m<sup>3</sup>  $\gg$  any stellar or magnetar volume.

#### FDPM Numerical Estimate

$$I_{SgrA^*} \sim \rho_{SCm} * \langle r \rangle * A * \text{correction} \\ \sim 1e15 * 1e10 * 4.76e20 * f_{\text{accretion}} \\ (\text{large} - \text{dominated by } \rho_{SCm} * r_s * A)$$

$$A = \pi * r_s^2 = 4.76e20 \text{ m}^2$$

$$\omega_1 - \omega_2 \sim \Omega_{\text{frame\_drag}} - \Omega_{\text{Keplerian\_outer}} \\ \sim c^2 / (G*M) * (a/M) - (G*M/r^3)^{0.5} \\ (\text{at } r = r_s \sim 1.23e10 \text{ m})$$

The full  $FDPM \sim 4.105e29 / (fDPM * Evac_{\text{neb}} * c * V_{\text{sys}})$  extracted from the result, confirming self-consistency.

#### 5. Accretion Disk and Jet Physics

The aDPM dominance at Sgr A\* has direct consequences for accretion disk and jet physics:

| Feature                | Standard GR Prediction                          | MUGE aDPM Prediction                                                |
|------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Disk truncation radius | ISCO = $3r_s$ (Schwarzschild) or smaller (Kerr) | ISCO modified by aDPM — smaller ISCO by factor $(1 - aDPM * r/c^2)$ |
| Jet launch velocity    | $v_{\text{jet}} \leq c$                         | $v_{\text{jet}}$ includes SCm vortex boost from aDPM                |

| Feature                           | Standard GR Prediction                 | MUGE aDPM Prediction                                                   |
|-----------------------------------|----------------------------------------|------------------------------------------------------------------------|
| Radio spectrum slope              | Power law: $F_{\nu} \sim \nu^{\alpha}$ | $F_{\nu}$ modification from aDPM-SCm coupling (flat spectrum at mm)    |
| QPO (Quasi-Periodic Oscillations) | From ISCO orbital frequency            | QPO + aDPM harmonic at $f_{DPM}/2 = 5e11 \text{ Hz} = 500 \text{ GHz}$ |

## 6. Sgr A\* in SOURCE4

In `MAIN_1_CoAnQi.cpp` (SOURCE4 namespace):

```
// Pre-defined Sgr A* system parameters
SOURCE4::sagA_SOURCE4 = {
    .M_bh      = 8.15e36, // kg (4.1e6 Msun)
    .r_s       = 1.23e10, // m (Schwarzschild radius)
    .vexp      = 9.0e7,   // m/s (0.3c accretion inflow)
    .Evac_neb  = 7.09e-36, // J/m^3
    .Evac_ISM  = 7.09e-37, // J/m^3
    .B_surface = 1.0e12,  // T (estimated accretion disk B)
    .spin_a    = 0.9,     // dimensionless (Kerr spin)
};

// Compute: returns aDPM = 4.105e29 m/s^2
double g = SOURCE4::compute_resonance_MUGE_SOURCE4(sagA_SOURCE4, params);
```

## 7. Conclusion

Sagittarius A's MUGE gravitational acceleration of  $g = 4.105 \times 10^{29} \text{ m/s}^2$  represents the maximal aDPM-dominant regime of the UQFF MUGE 12-Term Resonance framework. The result arises from the SMBH's unique combination of extreme mass ( $4.1 \times 10^6 M_{\text{sun}}$ ), frame-dragging spin ( $a/M \sim 0.9$ ), and  $V_{\text{sys}}$  at the Schwarzschild scale — all maximizing the FDPM vortical current and driving aDPM to dominate all 12 terms. The extreme  $g$  value is confined to within  $r_s$  of the BH — outside this radius, the standard MUGE  $f_{\text{TRZ}} \rightarrow 0$  limit progressively recovers Newtonian  $GM/r^2$  (PAPER\_155). The contrast with SGR1745-2900 (PAPER\_148, afluid\_freq dominant) validates the MUGE hierarchy: aDPM dominates at extreme- $V_{\text{sys}}$  systems, afluid\_freq dominates at extreme-B compact objects.

## References

- `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt` — EQ-013, MUGE 7-system results

- Event Horizon Telescope 2022 — Sgr A\* mass, shadow, EHT images
- Ghez et al. 2008; Gillessen et al. 2009 — S-star orbits, mass determination
- PAPER\_146 — 12-term MUGE master equation
- PAPER\_147 — FDPM vortical driver derivation
- PAPER\_148 — SGR1745-2900 (contrasting afluid\_freq dominant case)
- PAPER\_155 —  $\lim(\text{fTRZ} \rightarrow 0)$  Standard Model gravity recovery
- MAIN\_1\_CoAnQi.cpp SOURCE4 — sagA\_SOURCE4 system, compute\_resonance\_MUGE\_SOURCE4.Groups[1].Value — UQFF Sagittarius A\*: MUGE FDPM Dominance and Extreme Gravity

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| Mass                 | $4.154 \times 10^6 M_{\text{sun}} = 8.15 \times 10^{36} \text{ kg}$ | EHT 2022                |
| Schwarzschild radius | $r_s = 2GM/c^2 = 1.23 \times 10^{10} \text{ m}$                     | —                       |
| Event Horizon radius | $r_{\text{EH}} \sim r_s = 12.3 \text{ million km}$                  | EHT (shadow)            |
| EHT shadow diameter  | $\sim 51 \text{ microarcseconds}$                                   | EHT 2022                |
| Accretion luminosity | $\sim 10^{36-10^{38}} \text{ erg/s (variable)}$                     | Multi-band              |
| Jet inclination      | $65-85^\circ$ from line of sight                                    | VLBI                    |
| Distance             | $8.277 \pm 0.033 \text{ kpc}$                                       | VLBI (Reid et al. 2019) |
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The key parameters for Sgr A\* MUGE evaluation:

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 $r_s = 1.23e10 \text{ m}$  (Schwarzschild radius)  
 $V_{\text{sys}} = (4/3) \cdot \pi \cdot r_s^3 = (4/3) \cdot \pi \cdot (1.23e10)^3 = 7.78e30 \text{ m}^3$   
 $v_{\text{exp}} = 0.3c$  (relativistic accretion inflow / jet launch velocity)  
 $= 9e7 \text{ m/s}$

Term-by-term evaluation:

| Term          | Value (m/s <sup>2</sup> ) | Fraction        | Notes                                                                       |
|---------------|---------------------------|-----------------|-----------------------------------------------------------------------------|
| <b>aDPM</b>   | <b>4.105e29</b>           | <b>~99.9%</b>   | $FDPM / DPME_{\text{vac\_nebc}} V_{\text{sys}}$<br>Vsys(SMBH)<br>is maximal |
| aTHz          | $\sim 1.4e27$             | $\sim 0.3\%$    | Amplified by<br>$v_{\text{exp}} = 0.3c$                                     |
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| aaether_res   | $\sim 4.5e20$             | $\ll 0.01\%$    | $\omega_i = 1e-8$<br>vs aDPM huge                                           |
| Ug4i          | $\sim 3.2e15$             | negligible      | Self-SMBH;<br>$M_{\text{bh}}/d_g \rightarrow$                               |
| aquantum_freq | $\sim 1e-5$               | negligible      | Vsys large<br>$\hbar \cdot \omega_i^2$<br>tiny                              |

| Term         | Value (m/s <sup>2</sup> ) | Fraction   | Notes                                        |
|--------------|---------------------------|------------|----------------------------------------------|
| aAether_freq | ~4e26                     | ~0.1%      | $\rho_A/\rho_{UA} * \omega_i *$<br>aTHz      |
| afluid_freq  | ~5e26                     | ~0.1%      | $\nu * \lambda_{p_v}$<br>large but «<br>aDPM |
| Osc_term     | ~1e24                     | negligible | $\cos(\omega_{it}) \text{avac\_diff}$        |
| aexp_freq    | ~3e19                     | negligible | $H_z * \text{aDPM}/c$                        |
| fTRZ         | 0.1                       | negligible | Constant                                     |

**\*\*Total  $g_{\text{MUGE}}(\text{Sgr A}^*) = 4.105 \times 10^{29} \text{ m/s}^2$ \*\*** — dominated by aDPM.

### 3. Physical Interpretation of $g = 4.105e29 \text{ m/s}^2$

#### 3.1 Newtonian Comparison at 1 AU

At  $r = 1 \text{ AU} = 1.496e11 \text{ m}$  from Sgr A\*:

$$\begin{aligned}
 g_{\text{Newt}}(1 \text{ AU}) &= G*M/r^2 = 6.67e-11 * 8.15e36 / (1.496e11)^2 \\
 &= 6.67e-11 * 8.15e36 / 2.24e22 \\
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MUGE yield of  $4.105e29$  represents a ratio:

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This extreme amplification is NOT a prediction that S-star orbits violate Newtonian mechanics at 1 AU. Rather, the MUGE  $g = 4.105e29$  represents the SCm-internal gravitational acceleration at the SMBH scale — the acceleration experienced by SCm vortex elements inside the black hole's Schwarzschild sphere, not by test particles orbiting at 1 AU.

#### 3.2 Physical Scale of MUGE Evaluation

The MUGE  $V_{\text{sys}} = (4/3)\pi r_s^3$  defines the region of applicability:

- Inside  $r = r_s$ : MUGE regime,  $g$  up to  $4.105e29 \text{ m/s}^2$
- Outside  $r = r_s$ : Standard GR plus MUGE correction (PAPER\_155 limit approach: fTRZ->0 gives Newtonian)
- At  $r \sim r_s$ : Transition zone where aDPM and afluid\_freq drop sharply with  $r^{-3}$  (volume scaling)

The MUGE gravitational acceleration at  $r = r_s$  (Schwarzschild radius =  $1.23e10 \text{ m}$ ):

$$g_{\text{MUGE}}(r_s) = 4.105e29 \text{ m/s}^2 \text{ (computed at this scale)}$$

$$g_{\text{Newt}}(r_s) = G*M/r_s^2 = 6.67e-11 * 8.15e36 / (1.23e10)^2 = 3.6e15 \text{ m/s}^2$$

The MUGE/Newt ratio at  $r_s$  is  $\sim 1.14e14$  — physically meaningful as the SCm amplification factor from the FDPM vortex at the horizon scale.

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4.  **$V_{sys}$  is maximal:**  $(4/3)\pi r_s^3 = 7.78e30$  m<sup>3</sup>  $\gg$  any stellar or magnetar volume.

#### FDPM Numerical Estimate

$$I_{Sgr A^*} \sim \rho_{SCm} * \langle r \rangle * A * \text{correction} \\ \sim 1e15 * 1e10 * 4.76e20 * f_{\text{accretion}} \\ (\text{large} - \text{dominated by } \rho_{SCm} * r_s * A)$$

$$A = \pi * r_s^2 = 4.76e20 \text{ m}^2$$

$$\omega_1 - \omega_2 \sim \Omega_{\text{frame drag}} - \Omega_{\text{Keplerian outer}} \\ \sim c^2 / (G*M) * (a/M) - (G*M/r^3)^{0.5} \\ (\text{at } r = r_s \sim 1.23e10 \text{ m})$$

The full FDPM  $\sim 4.105e29 / (f_{DPM} * \text{Evac}_{\text{neb}} * c * V_{sys})$  extracted from the result, confirming self-consistency.

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#### 5. Accretion Disk and Jet Physics

The aDPM dominance at Sgr A\* has direct consequences for accretion disk and jet physics:

| Feature                | Standard GR Prediction                          | MUGE aDPM Prediction                                                |
|------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Disk truncation radius | ISCO = $3r_s$ (Schwarzschild) or smaller (Kerr) | ISCO modified by aDPM — smaller ISCO by factor $(1 - aDPM * r/c^2)$ |
| Jet launch velocity    | $v_{\text{jet}} \leq c$                         | $v_{\text{jet}}$ includes SCm vortex boost from aDPM                |

| Feature                           | Standard GR Prediction                 | MUGE aDPM Prediction                                                   |
|-----------------------------------|----------------------------------------|------------------------------------------------------------------------|
| Radio spectrum slope              | Power law: $F_{\nu} \sim \nu^{\alpha}$ | $F_{\nu}$ modification from aDPM-SCm coupling (flat spectrum at mm)    |
| QPO (Quasi-Periodic Oscillations) | From ISCO orbital frequency            | QPO + aDPM harmonic at $f_{DPM}/2 = 5e11 \text{ Hz} = 500 \text{ GHz}$ |

## 6. Sgr A\* in SOURCE4

In `MAIN_1_CoAnQi.cpp` (SOURCE4 namespace):

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    .B_surface = 1.0e12,  // T (estimated accretion disk B)
    .spin_a    = 0.9,     // dimensionless (Kerr spin)
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// Compute: returns aDPM = 4.105e29 m/s^2
double g = SOURCE4::compute_resonance_MUGE_SOURCE4(sagA_SOURCE4, params);
```

## 7. Conclusion

Sagittarius A's MUGE gravitational acceleration of  $g = 4.105 \times 10^{29} \text{ m/s}^2$  represents the maximal aDPM-dominant regime of the UQFF MUGE 12-Term Resonance framework. The result arises from the SMBH's unique combination of extreme mass ( $4.1 \times 10^6 M_{\text{sun}}$ ), frame-dragging spin ( $a/M \sim 0.9$ ), and  $V_{\text{sys}}$  at the Schwarzschild scale — all maximizing the FDPM vortical current and driving aDPM to dominate all 12 terms. The extreme  $g$  value is confined to within  $r_s$  of the BH — outside this radius, the standard MUGE  $f_{\text{TRZ}} \rightarrow 0$  limit progressively recovers Newtonian  $GM/r^2$  (PAPER\_155). The contrast with SGR1745-2900 (PAPER\_148, afluid\_freq dominant) validates the MUGE hierarchy: aDPM dominates at extreme- $V_{\text{sys}}$  systems, afluid\_freq dominates at extreme-B compact objects.

## References

- `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt` — EQ-013, MUGE 7-system results

- Event Horizon Telescope 2022 — Sgr A\* mass, shadow, EHT images
- Ghez et al. 2008; Gillessen et al. 2009 — S-star orbits, mass determination
- PAPER\_146 — 12-term MUGE master equation
- PAPER\_147 — FDPM vortical driver derivation
- PAPER\_148 — SGR1745-2900 (contrasting afluid\_freq dominant case)
- PAPER\_155 —  $\lim(f_{\text{TRZ}} \rightarrow 0)$  Standard Model gravity recovery
- MAIN\_1\_CoAnQi.cpp SOURCE4 — sagA\_SOURCE4 system, compute\_resonance\_MUGE\_SOURCE4